Fiscal Year:	FY 2022	**Task Last Updated:**	FY 01/06/2022
PI Name:	Holets, Lesya Ph.D.		
Project Title:	The Impact of the Prolonged LEO on Male Reproductive Health and Fertility in Mice on the Bion-M2 Mission		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
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City:	Kansas City	**State:**	KS
Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology NNH18ZTT002N:Russian Bion-M2 Mission
Start Date:	10/01/2020	**End Date:**	10/26/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:	Bion-M2		
Flight Assignment:	NOTE: End date changed to 10/26/2021 due to P.I. change to L. Christenson. Original end date 09/30/2023. Information per F. Hernandez/NASA-ARC. (Ed. 12/2/2021)		
Key Personnel Changes/Previous PI:	NOTE: P.I. changed to L. Christenson effective 10/27/2021. For subsequent reporting, please see "The Impact of the Prolonged LEO on Male Reproductive Health and Fertility in Mice on the Bion-M2 Mission (P.I. - Christenson)". (Ed. 12/2/2021)		
COI Name (Institution):	Larson, Melissa Ph.D. (University of Kansas Medical Center Research Institute, Inc.)		
Grant/Contract No.:	80NSSC20K1184		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Bion-M1 flight was the first opportunity to directly determine whether long-term microgravity under space flight conditions negatively affects testicular function and spermatogenesis in rodents. We reported that adult male mice flown for 30 days in orbit on the Bion-M1 showed significant declines in testicular integrity compared to ground controls including degeneration and disorganization of spermatogenic cells, appearance of gaps in the epithelium, increased number of residual bodies in the lumen, premature release of spermatids, abnormal Leydig cell morphology, and significant changes in expression of important markers of spermatogenesis and steroidogenesis. Prior published data demonstrates that long-term ground-based microgravity models using rodent hindlimb unloading (>3 weeks) show dramatic testicular atrophy and loss of spermatogenesis, leading to sterility. Microgravity, cosmic radiation, and stress may also interfere with sperm production during prolonged space flight. The Bion-M2 mission gives a unique opportunity to confirm and expand the significant changes in testis histopathology, sperm production and storage, and immune and endocrine regulatory dysfunction during long-term space flight. The testable hypothesis driving the experiments in this proposal is: Exposure to microgravity for 30-45 days causes disruption of spermatogenesis and steroidogenesis in adult mice leading to infertility. This research will harvest fresh testes and epididymides tissue from male mice to analyze post-flight: 1) testis and epididymis histopathology, cell apoptosis, immunohistochemical localization of key markers of spermatogenesis, steroidogenesis, and inflammatory signaling; 2) sperm motility of epididymal sperm and sperm DNA damage; 3) determination and analysis of genes and metabolic pathways altered in mice testis during long-term space flight; 4) Post-flight analysis will also incorporate mating trials and in vitro fertilization of surviving mice given a post-flight recovery period. As indicated in this NASA Research Announcement (NRA), our proposed project meets the research emphasis: 1) Physiology of the mouse endocrine function, and 6) Effects of radiation on organism health, biological function, and genetics. This project will fit within the design constraints of the Russian experiments on a non- interference basis. Collaborations with Russian investigators for post-flight recovery procedures such as mating trials will be explored for the Phase II application.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Achieving the specific aims will address critical knowledge gaps regarding the potential for multigenerational survival under combined spaceflight and increased radiation exposure as highlighted in the NASA Decadal Survey. Proof of the central hypothesis that long-term spaceflight has a negative impact on male reproductive health in mammals, will demonstrate that the reproductive system is at risk in long-term spaceflight, including: decline of spermatogenesis and sperm damage, Leydig cell damage that leads to reduction in testosterone levels and may negatively impact the health of offspring fathered post-flight.
Task Progress:	NOTE: P.I. changed to L. Christenson effective 10/27/2021. For subsequent reporting, please see "The Impact of the Prolonged LEO on Male Reproductive Health and Fertility in Mice on the Bion-M2 Mission (P.I. - Christenson)". (Ed. 12/2/2021)
Bibliography Type:	Description: (Last Updated:)